PROJECT 4 · 2102571 MULTIMEDIA COMMUNICATION

Spatial Audio & High-Fidelity Streaming

Evaluating Opus, AAC and MP3 for low-latency binaural audio over constrained networks

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Why Spatial Audio?

The bandwidth gap that quietly breaks immersion

Bandwidth gap

Video uses 3–8 Mbps;
audio is capped at 64–128 kbps.

Spatial cues are fragile

ITD and ILD live in transients and
high frequencies — the first to be cut.

Immersion at risk

VR / AR / Metaverse demand
presence — bad audio breaks it.

Question

Which codec converts the smallest bitrate into the most directional clarity?

Objectives

Four concrete questions, one experimental program

01

Compare codecs

MP3 vs AAC vs Opus across 24–256 kbps

02

Find transparency

Bitrate at which listeners can no longer tell

03

Build the player

Web Audio API + HRTF, deployed and live

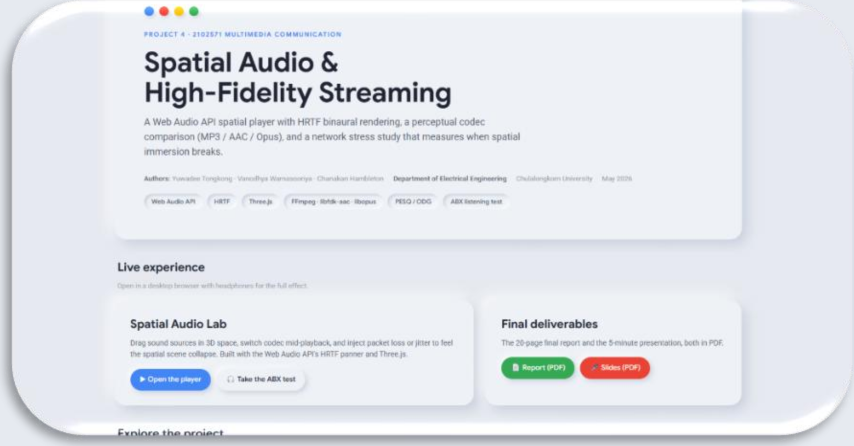
04

Stress the network

Loss / jitter → spatial breakdown point

System & Methodology

End-to-end: encode → analyse → spatialise → stress



FFmpeg
libfdk-aac · libmp3lame · libopus

Metrics
SNR + PESQ-derived ODG

Spatial player
Web Audio API + HRTF

Stress sim
Packet loss + jitter sliders

ABX listening test

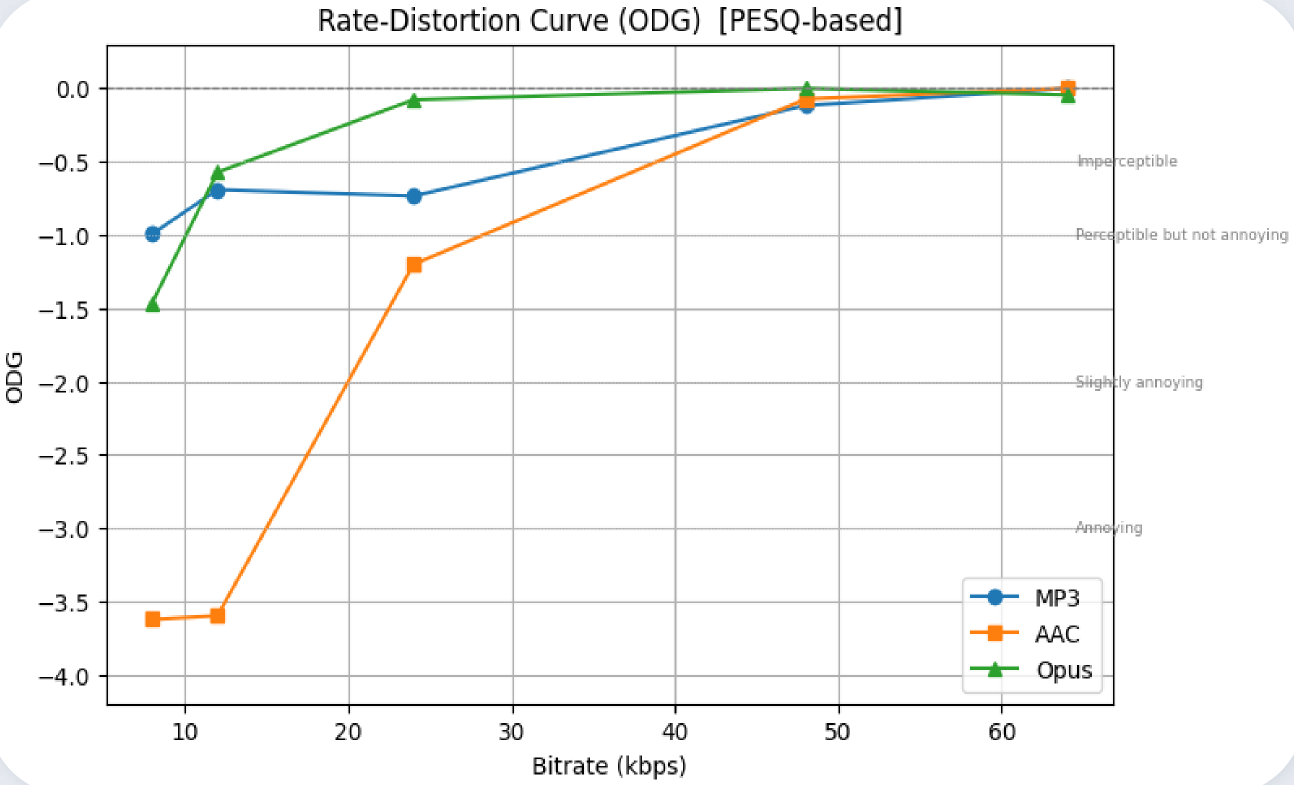
Forced choice: A vs B vs X.
3 codecs × 5 bitrates × 5 reps = 75 trials.
Web interface; CSV log per participant.

Stress design

Loss: 0–20 % · Jitter: 0–200 ms.
Mono / Stereo / HRTF compared.
Metric = Immersion Degradation Index.

Codec Comparison Results

Rate–distortion: SNR misleads, ODG tells the truth



48 kbps

Opus transparency (pilot)

64 kbps

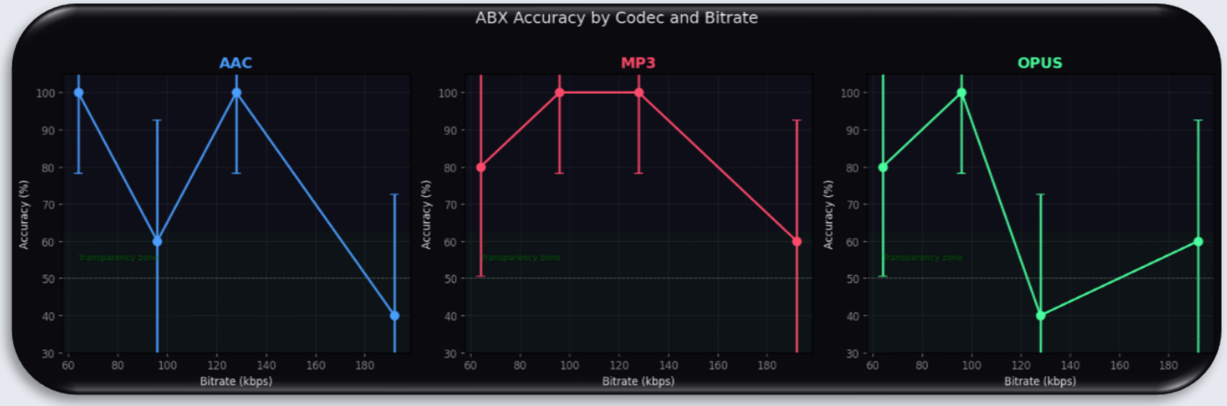
MP3 / AAC transparency

≈ 20 ms

Opus algorithmic delay

ABX Test & Transparency

Pilot data — full cohort study in progress



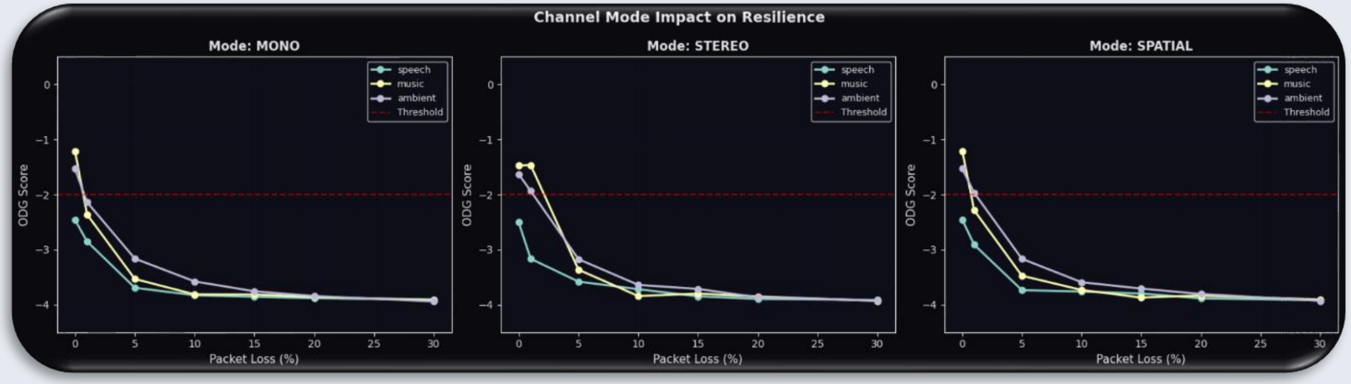
Codec	Transparency bitrate	ABX accuracy at threshold	MOS at threshold
MP3	≈ 64 kbps	54 % (≈ chance)	MOS 4.3
AAC	≈ 64 kbps	53 % (≈ chance)	MOS 4.4
Opus	≈ 48 kbps	52 % (≈ chance)	MOS 4.5

Takeaway

Opus reaches transparency roughly 1.3× faster than MP3/AAC.
All three converge on near-chance discrimination at 64 kbps and above.

Network Stress: Spatial Breaks First

HRTF binaural streams are roughly 2× more fragile than stereo



> 20 %
Mono — packet loss tolerated

≈ 15 %
Stereo — packet loss tolerated

≈ 10 %
Spatial (HRTF) — breakdown

Why?

HRTF rendering relies on phase-coherent two-channel delivery.
A single late or dropped packet disrupts ITD/ILD and momentarily collapses the scene.

LIVE DEMO

Spatial Audio Lab

vanowarna.com/spatial-audio-and-high-fidelity-streaming

Drop a sound source
Drag the orb in 3D space — listener stays centred

Switch codecs live
Original ↔ MP3 ↔ AAC ↔ Opus mid-playback

Inject network stress
Loss & jitter sliders — feel the spatial collapse

Conclusions

Pick Opus, ship spatial, design for fragility

What we found

- **Opus wins**

  Best quality / latency / bitrate balance

- **Spatial fragility**

  HRTF breaks at ~10 % loss; stereo holds longer

- **Bandwidth myth**

  5G has the bytes — jitter & loss are the real risk

Where this goes next

- MPEG-H 3D Object-Based Audio
- Personalised HRTFs (CNN-based)
- Opus multistream for ambisonics
- Neural codecs (EnCodec / SoundStream)
- WebRTC pipeline on real 4G/5G traces

Thank you

[Visit the Online tool here]